Watch Your Teacher Build an AI

Name _____

Work with a partner. Compare answers when you finish each problem.

1. How long by hand? 140 cards, 300 passes, 1 minute per card.

140 $\times$ 300 = _____ cards = _____ minutes $\approx$ _____ hours $\approx$ _____ days

2. Six steps to build an AI. Number the steps in order, 1 to 6.

	Train: many passes, fix the weights		Get data
	Split the data: training and test		Test, then use it
	Ask a question		Build the network

Why does the network never train on the test data? _____

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3. Predict, train, and test. Write your prediction **before** each run.

Run	Question	My prediction (test %)	Training %	Test %
1	Fighting or Psychic? (2 numbers)			
2	Fighting or Psychic? (8 numbers)			
3	All 18 types			

Word bank: data overfitting

4. What did you see?

In Run 2, the training score went up, but the test score didn't. The network learned the training examples by heart. This is called _____.

In Run 3, the network wasn't much better than guessing. Why? _____

An AI can only learn what is in its _____.

5. Talk about it. What if someone used an AI like this to decide who gets a job?

A problem could be _____

_____

_____

Go Further (Day 2). Ask the network. Try 3 Pokémon you know.

Pokémon	The network said	Really	Right?	Test Pokémon?